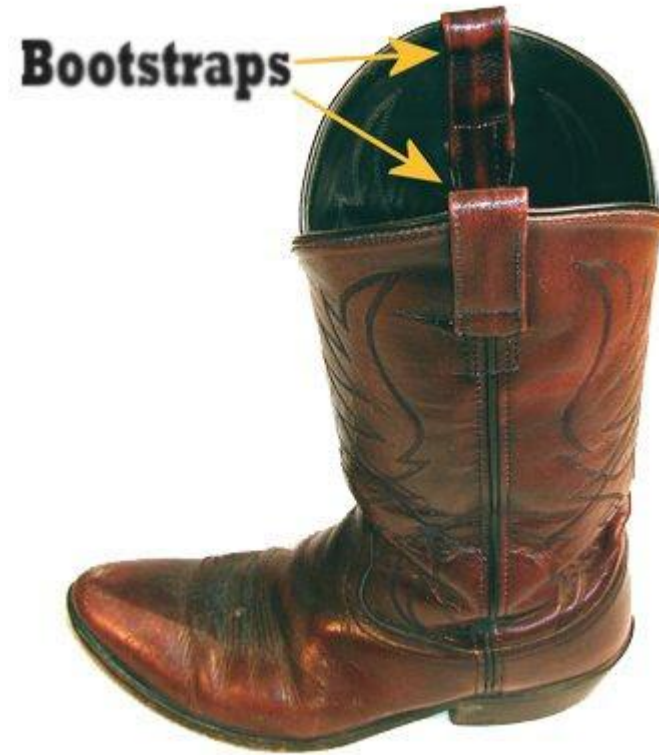


The bootstrap



Overview

Review of confidence intervals

Review and continuation of the bootstrap

Calculating bootstrap confidence intervals in R

If there is time: more practice creating confidence intervals

Announcement

Homework 5 has been posted!

It is due on Gradescope on **Sunday October 11th at 11pm**

- **Be sure to mark each question on Gradescope!**

Quick review of confidence intervals

Review: confidence intervals

A **confidence interval** is an interval computed by a method that will contain the *parameter* a specified percent of times

The **confidence level** is the percent of all intervals that contain the parameter

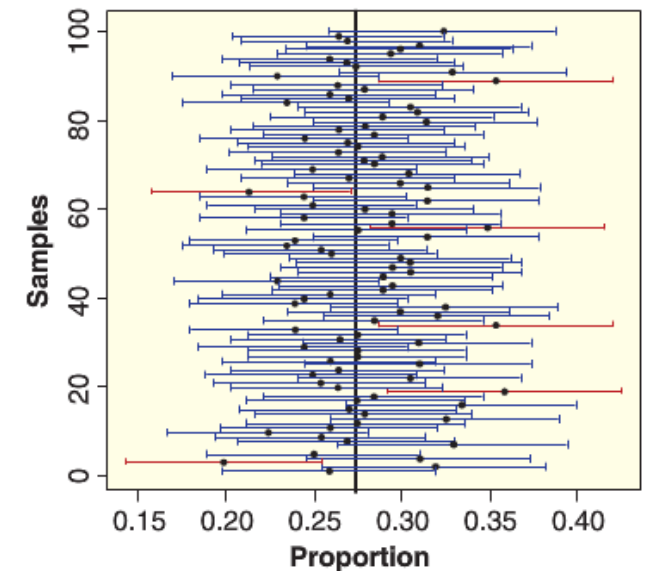
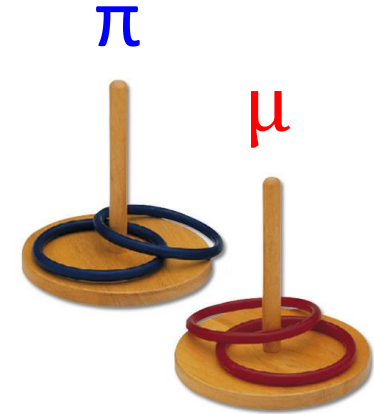


There is a tradeoff between:

- The **confidence level**
- The **confidence interval size**



"Parameter catchers"



Example: Quinnipiac poll

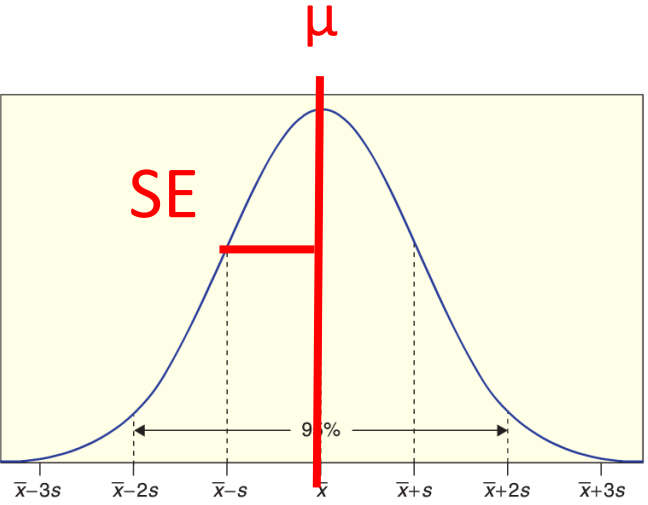
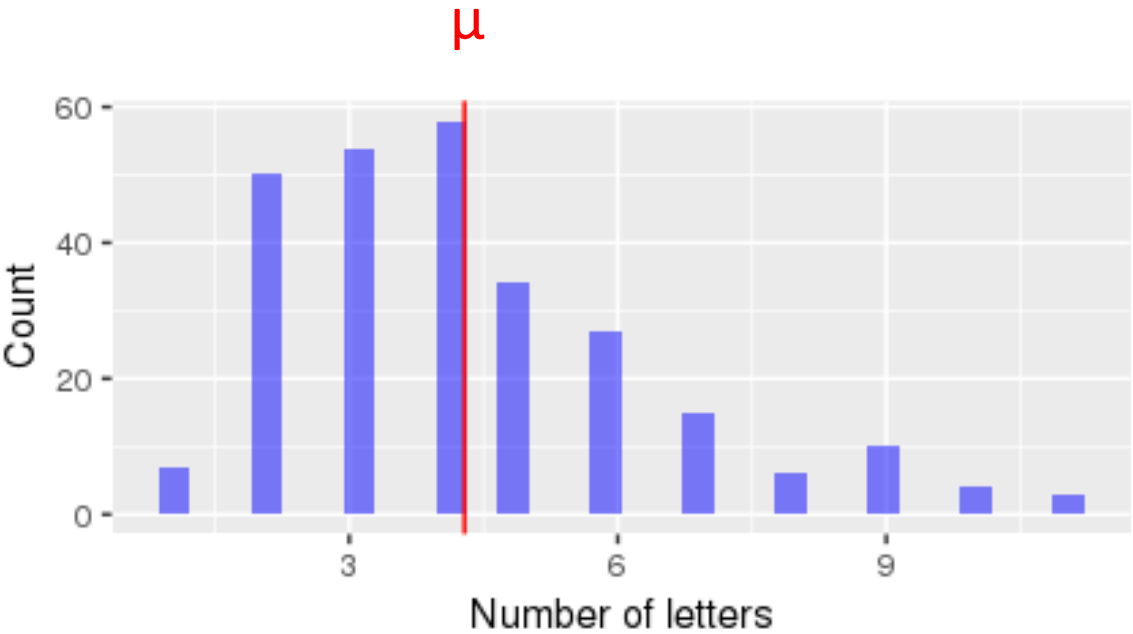
A [recent Quinnipiac poll](#) surveyed 1,202 US Adults and found that 53% believe AI will do more harm than good in their day-to-day lives with a margin of error of 3.5%

How do we interpret this?

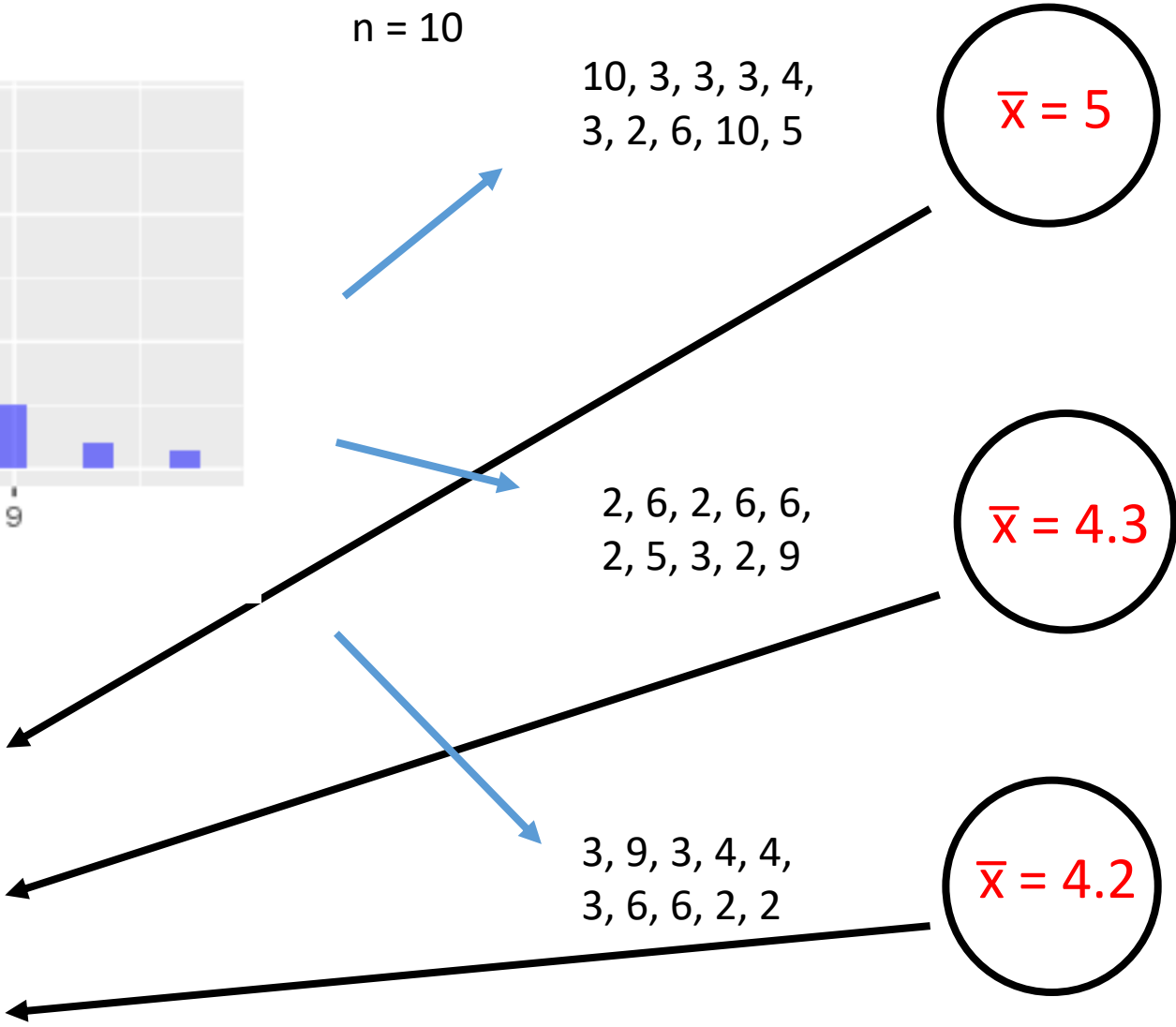
The *population **parameter*** (π) lies somewhere between 49.5%. to 56.5%

I.e., if they sampled all Americans, the true population proportion (π) of people who believe AI will do more harm than good in their day-to-day lives is in this range

Review: sampling distribution illustration



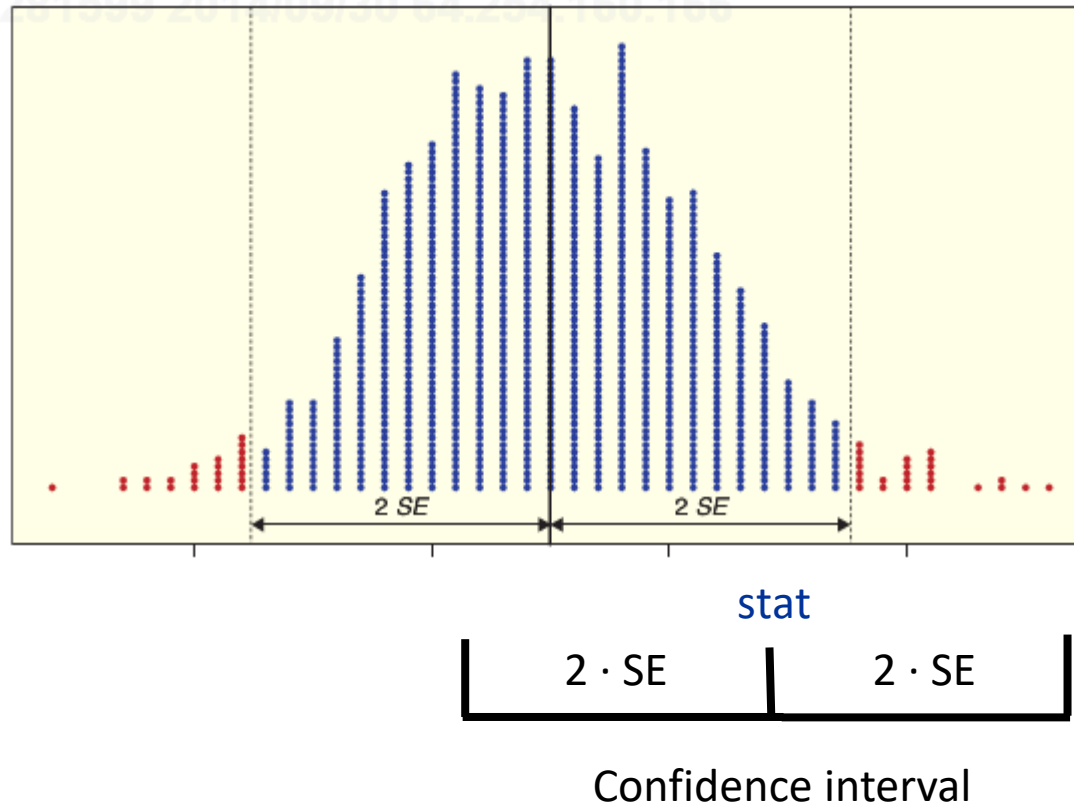
Sampling distribution!



Sampling distributions

Q: For a sampling distribution that is a normal distribution, what percentage of **statistics** lie within 2 standard deviations (SE) for the population mean?

A: 95%



If we had:

- A statistic value
- The SE

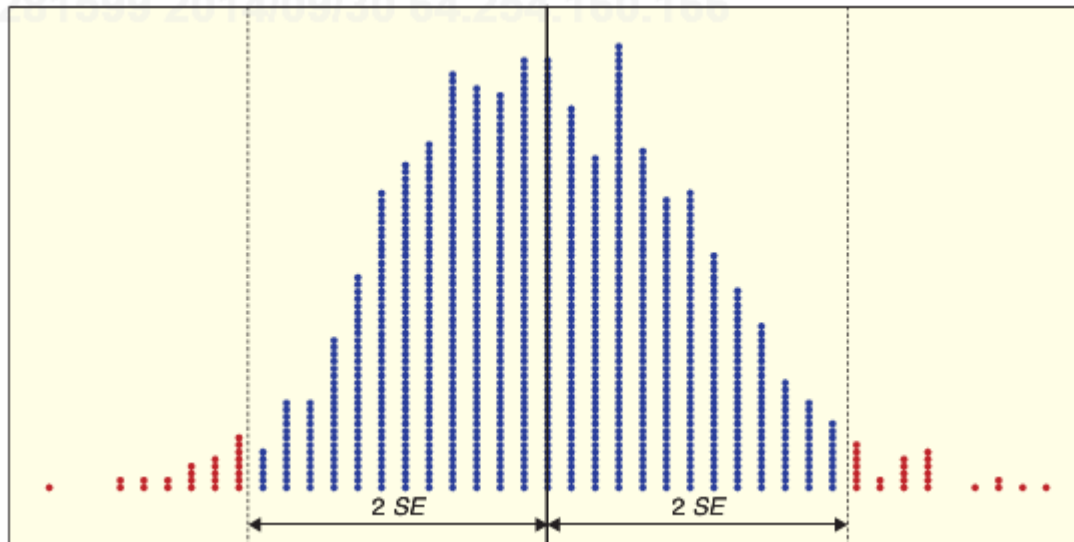
We could compute a 95% confidence interval!

$$CI_{95} = \text{stat} \pm 2 \cdot SE$$

Sampling distributions

Q: For a sampling distribution that is a normal distribution, what percentage of **statistics** lie within 2 standard deviations (SE) for the population mean?

A: 95%



Confidence interval

If we had:

- A statistic value
- The SE

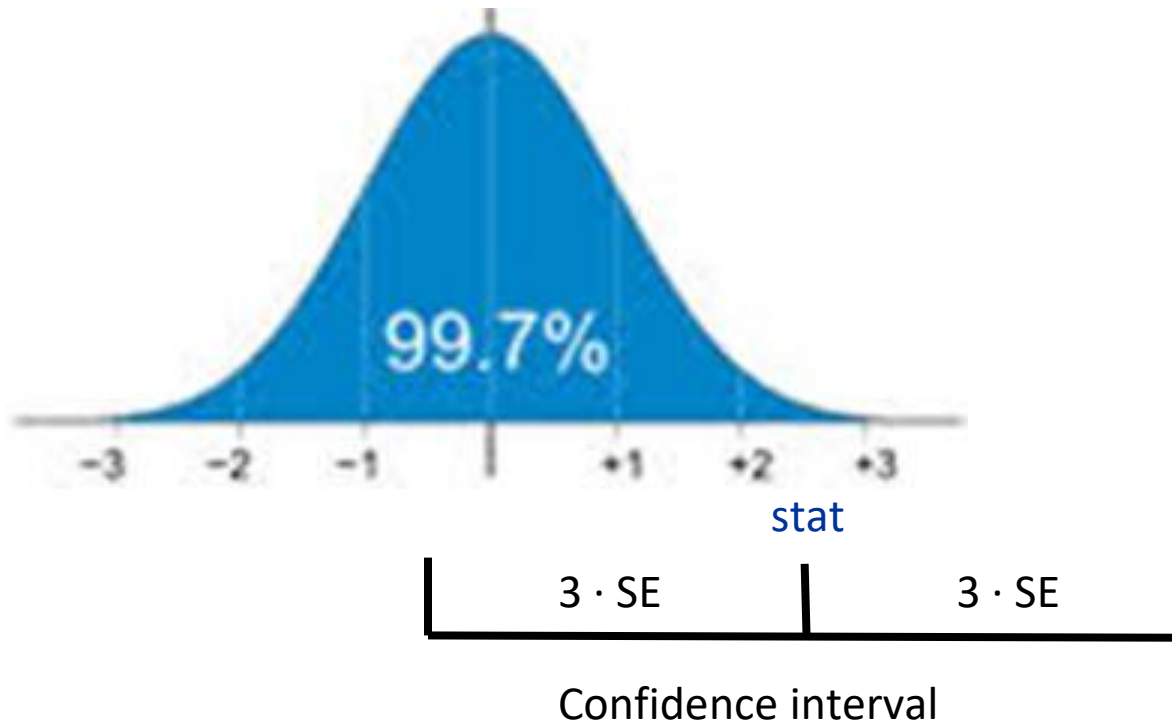
We could compute a 95% confidence interval!

$$CI_{95} = \text{stat} \pm 2 \cdot SE$$

Confidence intervals for other confidence levels

Q: How could we get a 99.7% confidence interval confidence level?

A: For normally distributed statistics, 99.7% of our statistics lie within 3 standard deviations (i.e., standard errors) of the mean

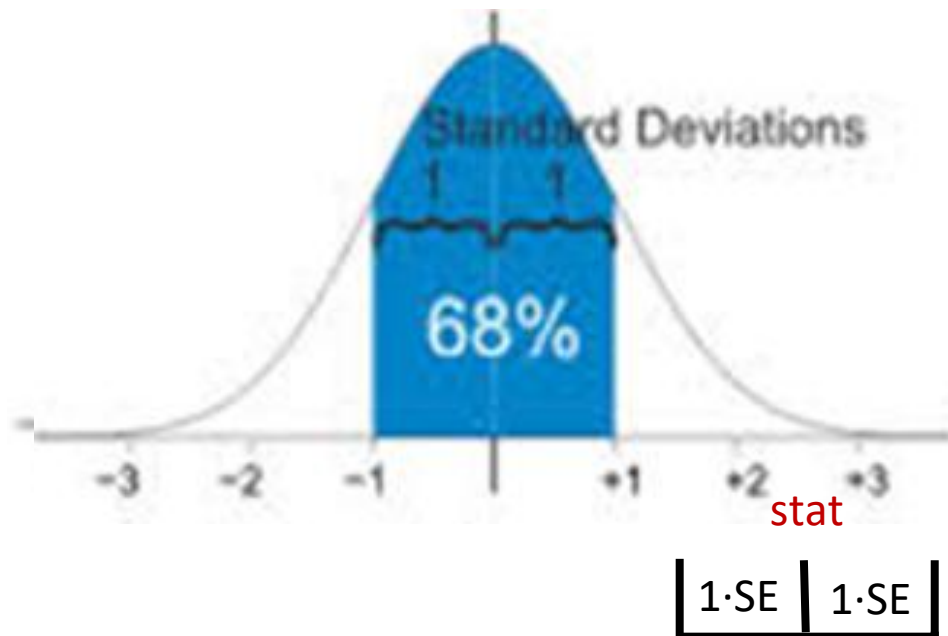


$$CI_{99.7} = \text{stat} \pm 3 \cdot SE$$

Confidence intervals for other confidence levels

Q: How could we get a 68% confidence interval confidence level?

A: For normally distributed statistics, 68% of our statistics lie within 1 standard deviation (i.e., standard error) of the mean



$$CI_{99.7} = \text{stat} \pm 3 \cdot SE$$

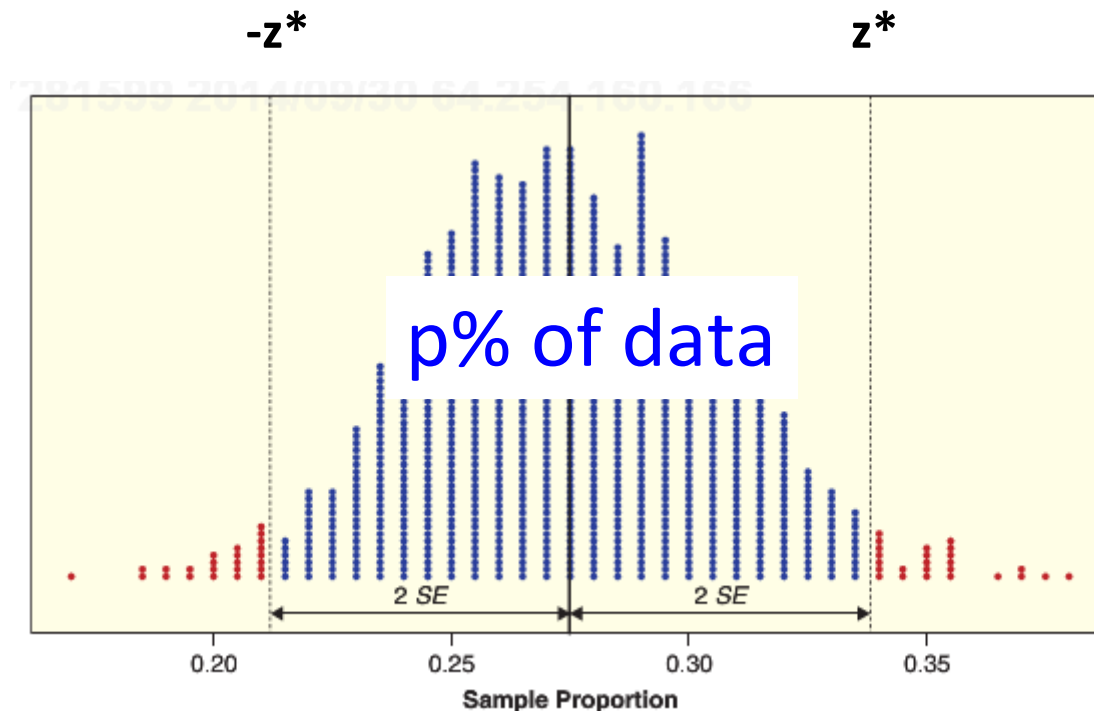
$$CI_{68} = \text{stat} \pm 1 \cdot SE$$

Confidence interval

Confidence intervals for other confidence levels

Q: How could we get a confidence interval for the q^{th} confidence level?

A: We need to find the **critical value** z^* such that $p\%$ of our statistics are within $\pm z^* \cdot \text{SE}$ for a normal distribution



$$CI = \text{stat} \pm z^* \cdot \text{SE}$$

Critical value z^*

Critical value

For a 95% CI, we choose $z^* = 2$

- 95% of our statistics lie within -2 and 2 SE of the parameter
- (1.96 to be more precise)

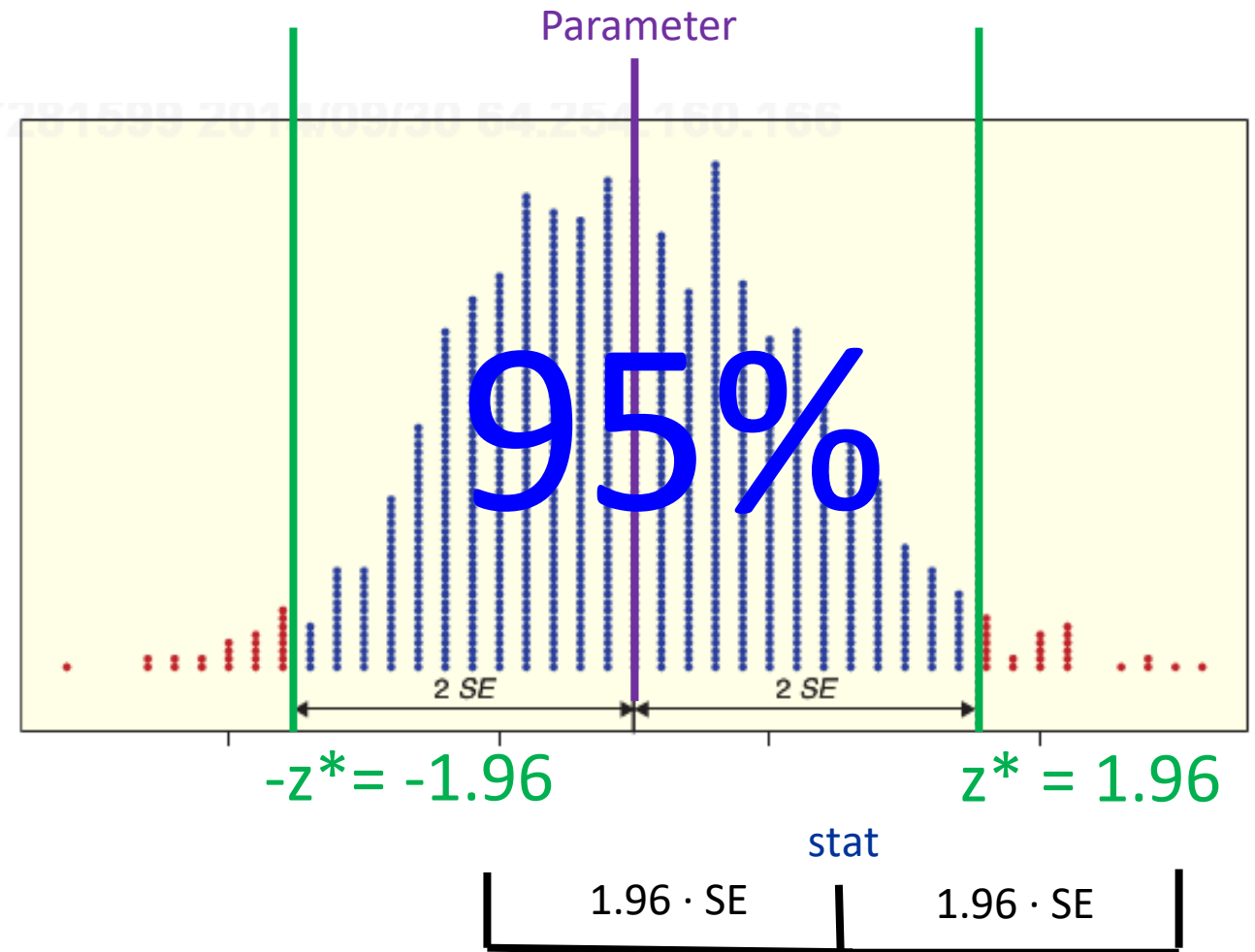
```
crit_val_95 <- cnorm(.95)
```

1.96

Critical value z^*

confidence interval: $\text{stat} \pm z^* \cdot \text{SE}$

$= \text{stat} \pm 1.96 \cdot \text{SE}$



Critical value

We can choose critical values q^* for other confidence levels

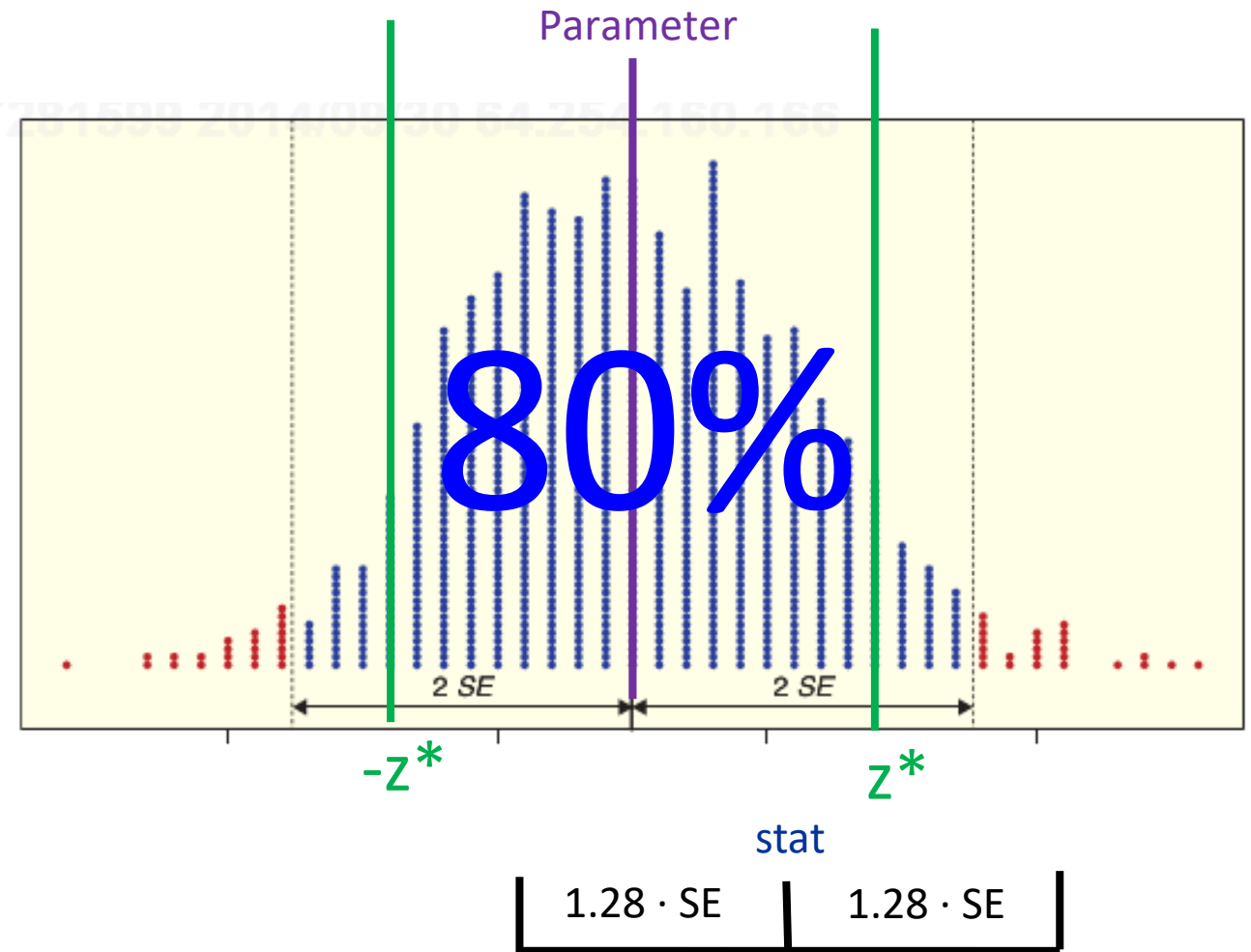
- E.g. for an 80% CI, we choose q^* so that 80% of our statistics lie within $-q^*$ and q^* SE of the parameter

```
crit_val_80 <- cnorm(.80)
```

1.28

Critical value z^*

confidence interval: $\text{stat} \pm z^* \cdot \text{SE}$



$$= \text{stat} \pm 1.28 \cdot \text{SE}$$

Sampling distributions

Q: Could we calculate the SE by repeatedly sampling from a population to create sampling distribution, and then take the sd of this sampling distribution?

- A: Not in the real world because it would require running our experiment over and over again...

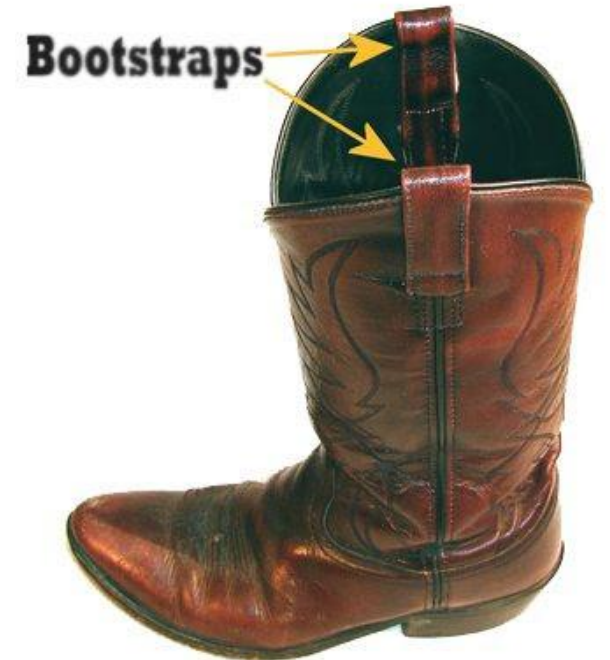


Sampling distributions

Q: If we can't calculate the sampling distribution, what else can we do?

- A: We can pick ourselves up from the bootstraps

1. Estimate SE with SE^* *from a single sample of data*
2. Then use $\bar{x} \pm 2 \cdot SE^*$ to get the 95% CI



Questions?



The bootstrap continued



Sampling distributions

As previously discussed, in practice we can't calculate the sampling distribution by repeating sampling from a population ☹️

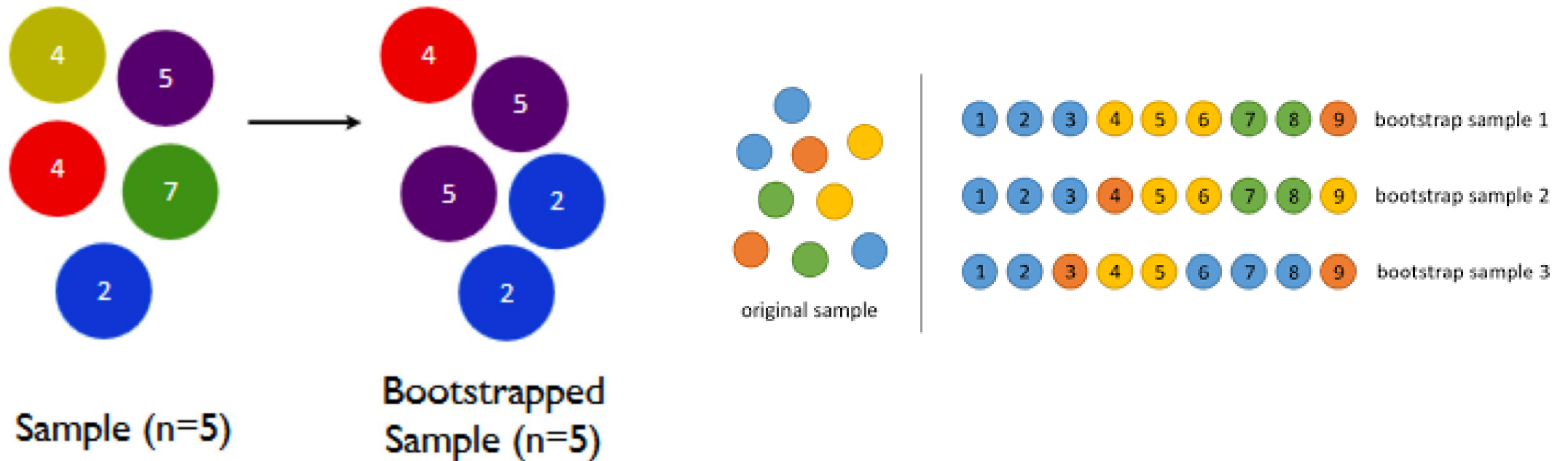
- Therefore, we can't get the SE from the sampling distribution ☹️

We have to pick ourselves up by the bootstraps!

1. Estimate SE with SE^*
2. Then use $\text{stat} \pm 2 \cdot SE^*$ to get the 95% CI

Sampling with replacement

Sampling data **with replacement** means we can select some of the values more than once, and might not select other values at all



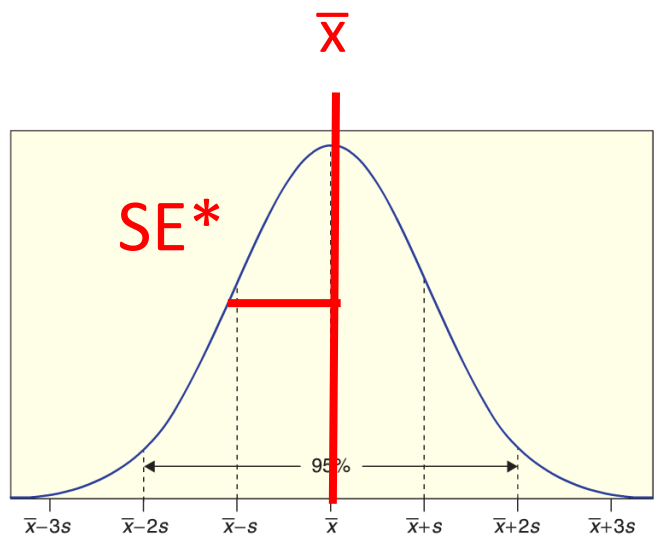
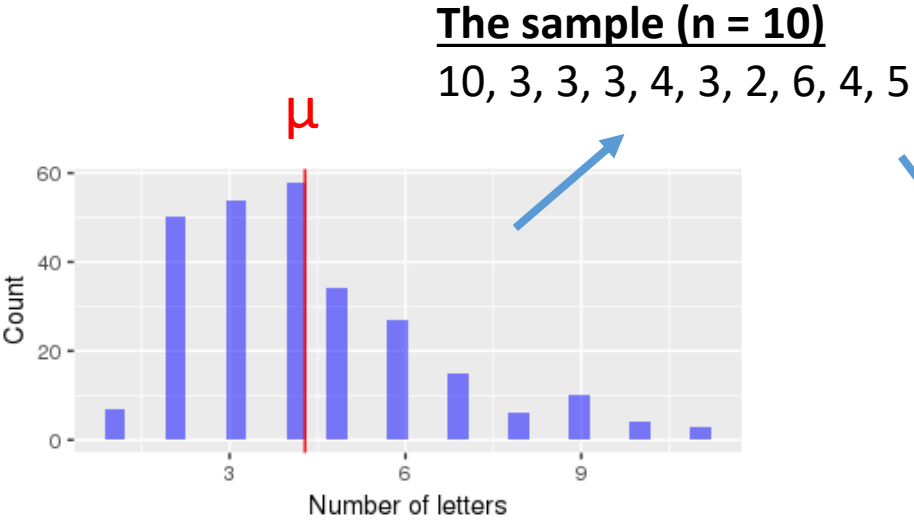
Plug-in principle

Suppose we get one sample of size n from a population

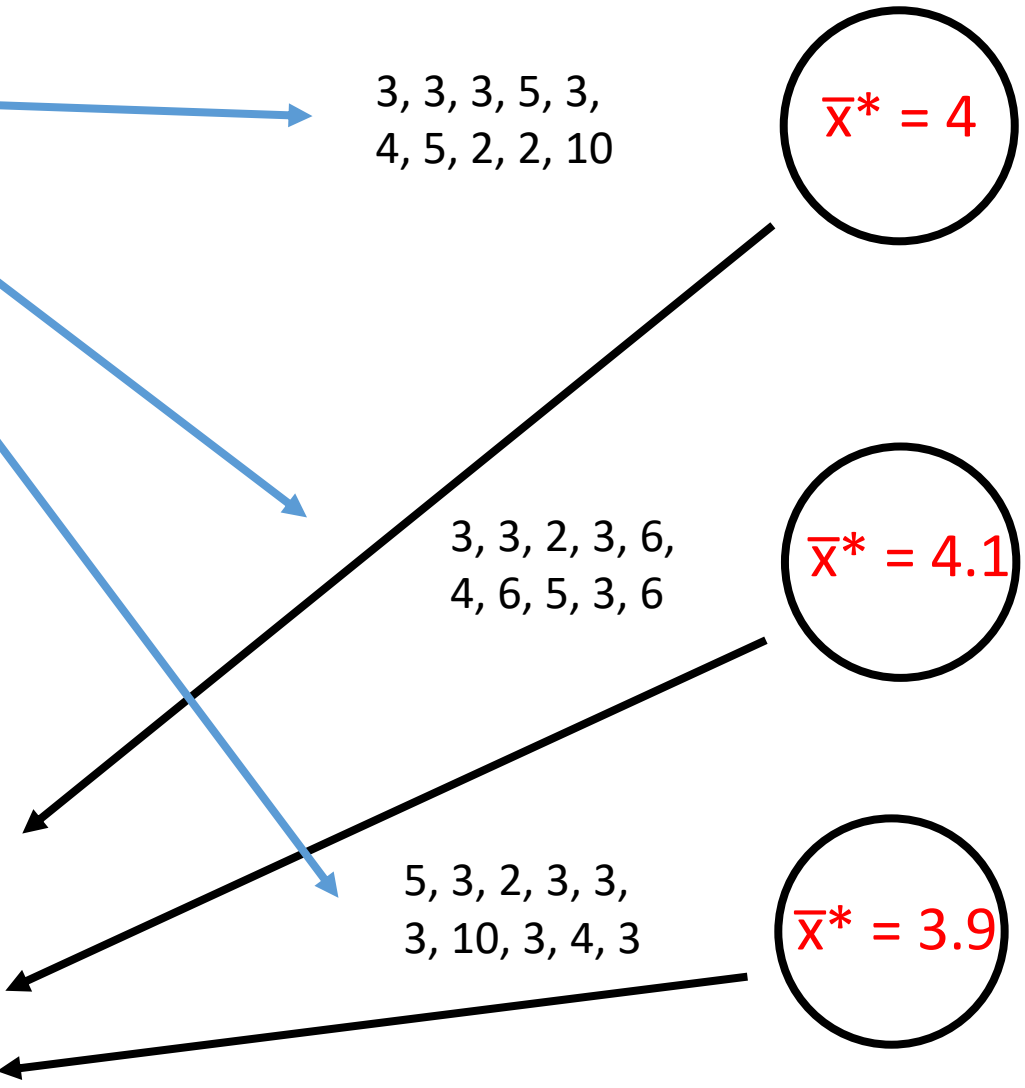
We ***pretend that this sample is the population*** (plug-in principle)

1. We then sample n points ***with replacement*** from ***our sample***, and compute our statistic of interest
2. We repeat this process 1000's of times and get a **bootstrap sample distribution**
3. The **standard deviation of this bootstrap distribution** (SE* bootstrap) is a good approximate for standard error SE from the real sampling distribution

Bootstrap distribution illustration



Bootstrap distribution!



Notice there is no 9's in the bootstrap samples

95% Confidence Intervals

When a bootstrap distribution for a sample statistic is approximately normal, we can estimate a 95% confidence interval using:

$$\text{Statistic} \pm 2 \cdot SE^*$$

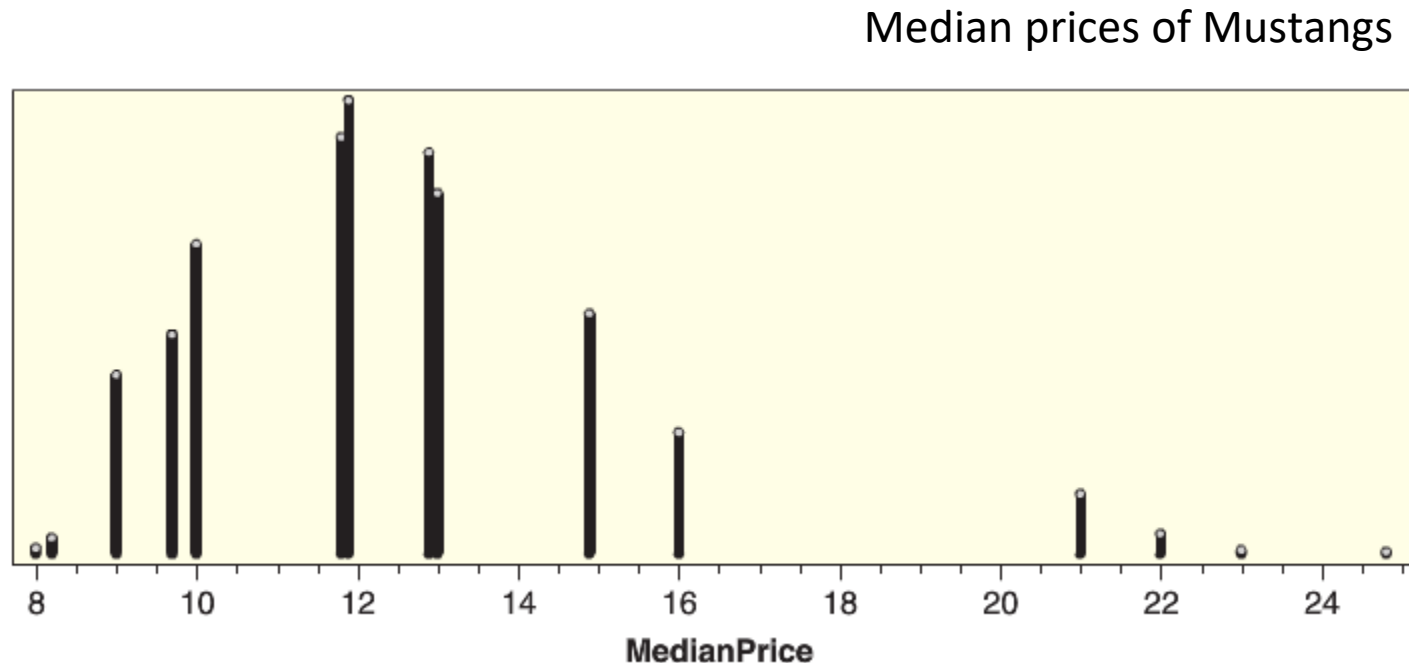
Where SE^* is the standard error estimated using the bootstrap

Findings CIs for many different parameters

The bootstrap method works for constructing confidence intervals for many different types of parameters!

Caution: the bootstrap does not always work

Always look at the bootstrap distribution, if it is poorly behaved (e.g., heavily skewed, has isolated clumps of values, etc.), you should not trust the intervals it produces.



Bootstrap confidence intervals in R

What are the steps needed to create a bootstrap SE?

1. Start with a sample

2. Repeat steps 10,000 times

- a. Resample the points in the sample to get a bootstrap sample
- b. Compute the statistic of interest on the bootstrap sample

3. Take the standard deviation of the bootstrap distribution to get SE*

Sampling with replacement from a vector

```
my_sample <- c(3, 1, 4, 1, 5, 9)
```

To get a sample of size n = 6 with replacement:

```
boot_sample <- sample(my_sample, 6, replace = TRUE)
```

Sampling distribution in R

```
my_sample <- c(21, 29, 25, 19, 24, 22, 25, 26, 25, 29)
```

```
bootstrap_dist <- do_it(10000) * {  
    curr_boot <- sample(my_sample , 10, replace = TRUE)  
    mean(curr_boot)  
}
```

```
SE_boot <- sd(bootstrap_dist)
```

Bootstrap confidence interval in R

```
obs_mean <- mean(my_sample)
```

```
CI_lower <- obs_mean - 2 * SE_boot
```

```
CI_upper <- obs_mean + 2 * SE_boot
```

Let's try it in R!

Practice creating confidence intervals in R

Example 1: One true love?

A survey asked 2625 people whether they agreed with the statement “There is only one true love for each person”

1812 of the respondents disagreed

Compute a 90% confidence interval for the proportion who **agree** that there is one true love



Example 2: Average body temperature

[Machowaik et al \(1992\)](#) recorded the body temperature of 130 observations people

They calculated a 95% confidence interval for the body temperatures to be: [98.12, 98.37]

Q: How do we interpret these results?

- A: The True mean body temperature of humans μ is (likely) between 98.12 and 98.37 degrees

Q: Is this what you would expect?

- A: No, I was told that the average human body temperature is $\mu = 98.6$ degrees
 - It turns out [that body temperatures have been decreasing](#)

Introduction to hypothesis tests

A quick note on probability

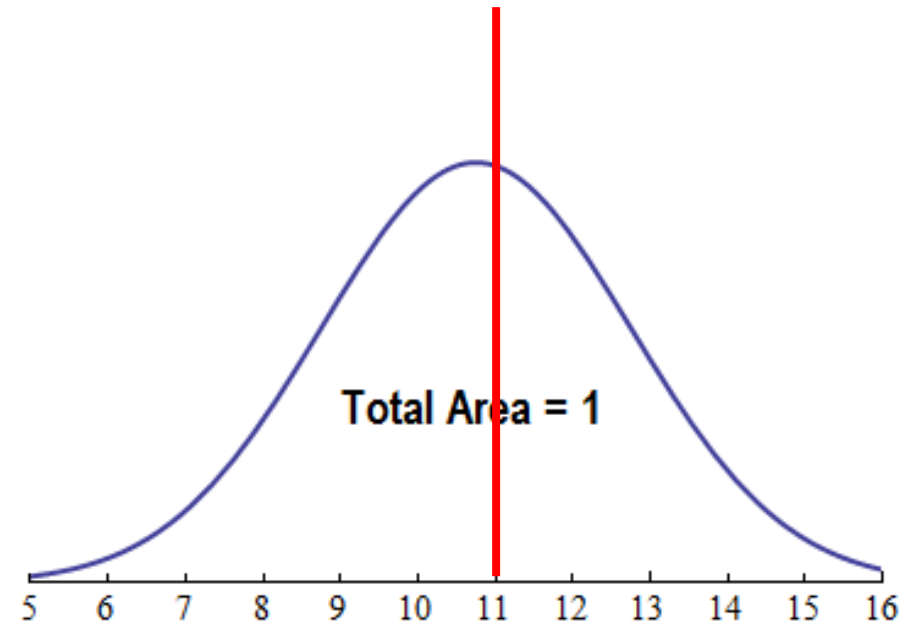
Probability is a way of measuring the likelihood that an event will occur

Probability models assigns a number between 0 and 1 to the outcome of an event (outcome) occurring

We can use a probability model to calculate the probability of an event

For example:

- $P(X < 11) = 0.55$
- $P(X > 20) = 0$



Statistical tests (hypothesis test)

A **statistical test** uses data from a sample to assess a claim about a population

Example 1: we might make the claim that Trump's approval rating for all US citizens is 42%

How can we write this using symbols?

- $\pi = .42$

Statistical tests (hypothesis test)

A **statistical test** uses data from a sample to assess a claim about a population.

Example 2: we might make the claim that the average height of a baseball player is 72 inches

How can we write this using symbols?

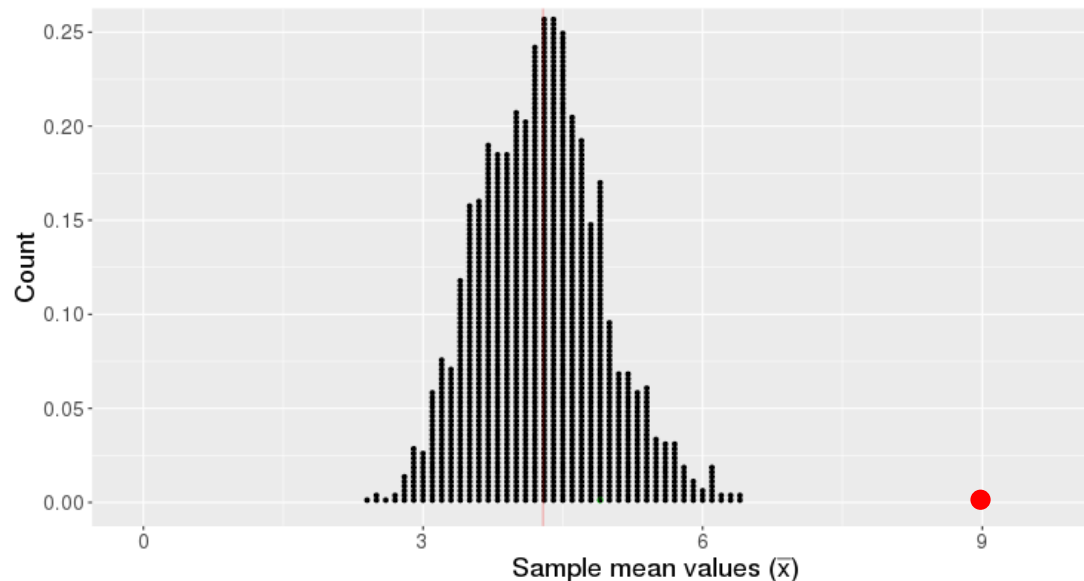
- $\mu = 72$

Basic hypothesis test logic

We start with a claim about a population parameter

- E.g., $\mu = 4$

This claim implies we should get a certain distribution of statistics



If our observed statistic is highly unlikely, we reject the claim

Are dolphins capable of abstract communication?

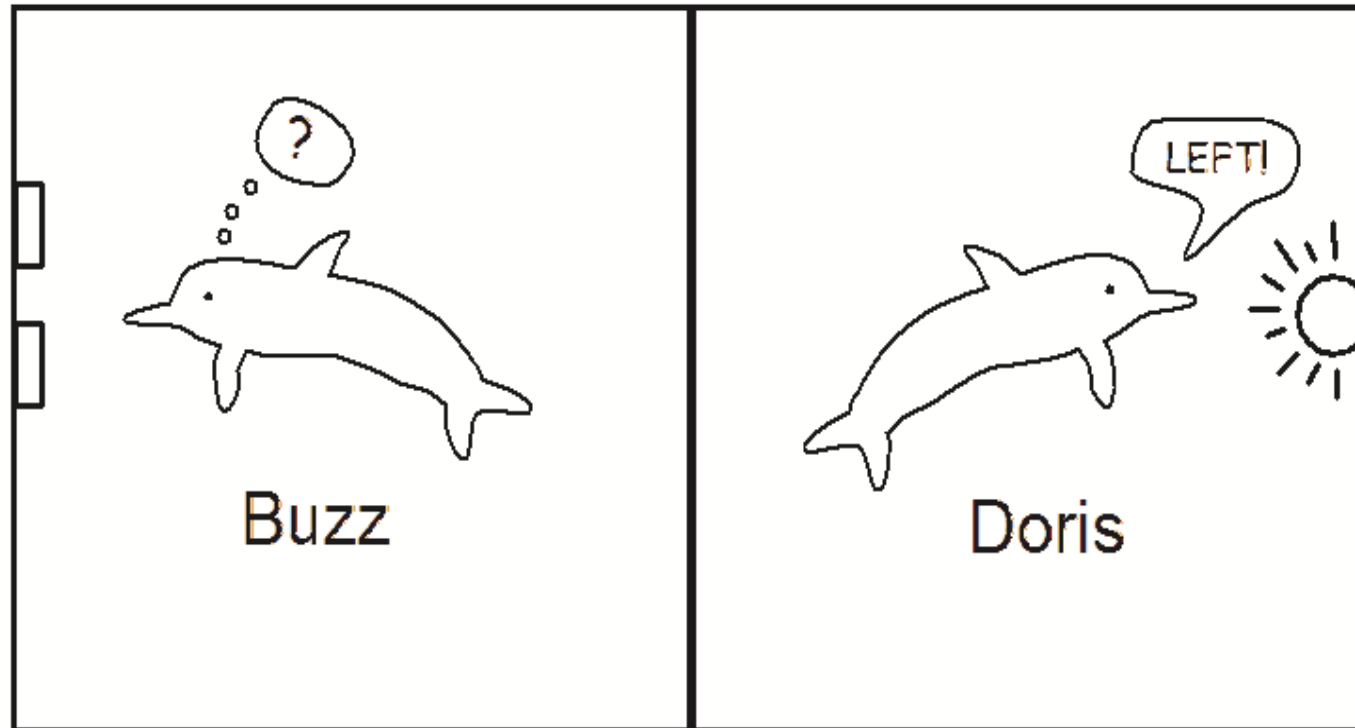
Dr. Jarvis Bastian in the 1960's wanted to know whether dolphins are capable of abstract communication

He used an old headlight to communicate with two dolphins (Doris and Buzz)

- Steady light = push button on right to get food
- Flashing = push button on the left to get food



A canvas was then put in the middle of the pool with Doris on one side and Buzz on the other



Left



Right



Buzz got 15 out of 16 trials correct

Please answer the following questions

1. What are the cases here?
2. What is the variable of interest and is it categorical or quantitative?
3. What is the observed statistic - and what symbols should we use to denote it?
4. What is the population parameter we are trying to estimate - and what symbol should we use to denote it?
5. Do you think the results are due to chance?
 - i.e., how many correct answers do you think Buzz would have gotten if he was guessing?
6. Are dolphins capable of abstract communication?

The dolphin communication study

7. If Buzz was just guessing, what would you expect the value of the parameter to be?

$$\pi = 0.5$$

8. If Buzz was not guessing, what would you expect the value of the parameter to be?

$$\pi > 0.5$$

Chance models

How can we assess whether 15 out of 16 correct trials ($\hat{p} = .975$) is beyond what we would expect to see by chance?

- i.e., beyond what we would expect to see if $\pi = 0.5$?

If buzz was guessing we can model his guesses as a coin flip:

Heads = correct guess

Tails = incorrect guess

Let's flip a coin 16 times and see how many times we get 15 heads

Chance models

To really be sure, how many repetitions of flipping a coin 16 times should we do?

Any ideas how to do this?

Flipping coins using SDS1000 functions

`rflip()` returns the number of heads out of `num_flip` coin flips:

```
rflip(num_flips, prob = .5)
```

num_flips: the number of times to flip a coin

- 16 for Doris/Buzz

prob: the probability of success on each trial

- .5 if Buzz was guessing

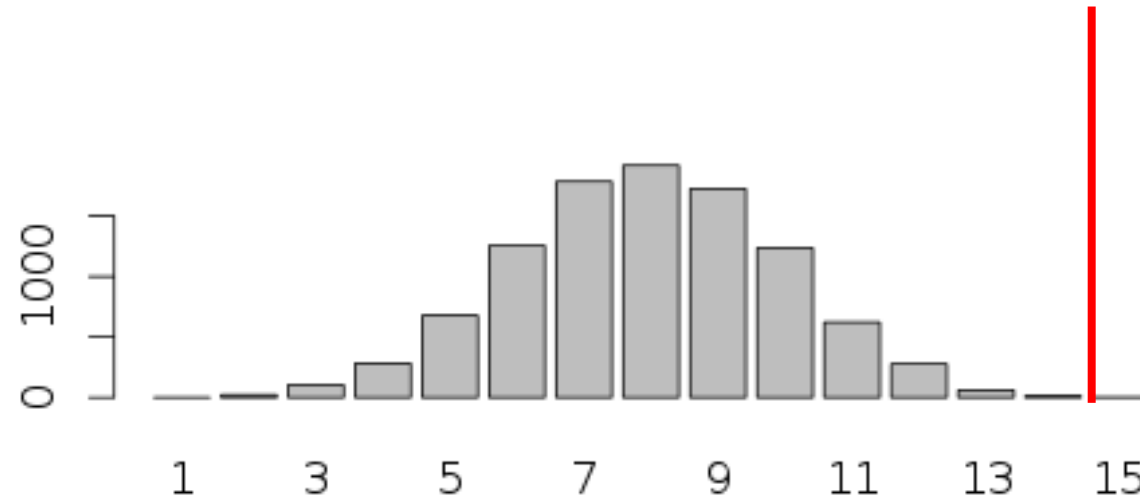
We can repeat flips many times using the `do_it()` function:

```
library(SDS1000)
```

```
flip_simulations <- do_it(10000) * {  
  rflip(16, prob = .5)  
}
```

Simulating Flipping 16 coins 10,000

0	0
1	1
2	22
3	105
4	283
5	679
6	1257
7	1786
8	1920
9	1726
10	1238
11	623
12	279
13	63
14	15
15	3
16	0



Q_{10} : Is it likely that Buzz was guessing?

Q_{11} : Are dolphins capable of abstract communication?

